

## 1. DFA M

### 1.a. Is $1011 \in L(M)$

This string is rejected, since  $q_8$  does not have a 1 transition into an accept state

### 1.b. Is $\langle M, 1011 \rangle \in A_{\text{DFA}}$

This combo of DFA and string is rejected since  $M$  does not accept 1011

### 1.c. Is $\langle M, 1010 \rangle \in A_{\text{DFA}}$

Yes, this combo of DFA and string is accepted, we follow the states  $q_0 \rightarrow q_2 \rightarrow q_5 \rightarrow q_8 \rightarrow q_5$

### 1.d. Is $\langle M \rangle \in E_{\text{DFA}}$

No,  $E_{\text{DFA}}$  rejects  $M$  since we have shown that  $M$  accepts atleast one string

## 2. Is $A$ decidable

We know that we can change a regular expression into a DFA, lets define  $R_{\text{DFA}}$  and  $S_{\text{DFA}}$  as the DFA's for  $R$  and  $S$ .

We create a new DFA,  $R'_{\text{DFA}}$  that is the complement of  $R_{\text{DFA}}$ , this DFA rejects all strings accepted by  $R_{\text{DFA}}$ . Now we create a final DFA  $R' S_{\text{DFA}}$  which corresponds to the intersection of  $R'_{\text{DFA}}$  and  $S_{\text{DFA}}$ . Finally we put our new  $R' S_{\text{DFA}}$  into  $E_{\text{DFA}}$ , if  $E_{\text{DFA}}$  accepts then we have shown that  $S$  is a sublanguage of  $R$  and is therefore decidable.

## 3. Non negative integer pairs

### 3.a. show that $W$ is both infinite and countable

We can show that  $W$  is infinite and countable by creating a function

$$f(a, b) = p^a q^b$$

where  $p$  and  $q$  are prime numbers where  $p \neq q$ . This is possible because every integer can be factorized into a unique product of primes (*see the unique factorization theorem*).

### 3.b. Show that $r(y) := S(y)$ is finite for all $y$

Given the rules of  $S(y)$  that  $x < y$  we have two possible situations:

$$x_2 < y_2$$

$x$  has  $x_2 + 1$  possible combos, in other words  $y_2$ , for where  $x_2$  can be any integer between 0 and  $y_2 - 1$ .

$$x_2 = y_2 \text{ and } x_1 < y_1$$

now  $x$  only has  $y_1$  possible combos since  $x_2$  does not change.

### 3.c. Closed form

For the rule  $x_2 < y_2$   $x_2$  can take any value between 1 and  $y_2 - 1$ ,  $x_1$  can take values between 0 and  $x_2 - 1$ , this can be shown by the closed form

$$\sum_{i=1}^{y_2-1} i = \frac{y_2(y_2-1)}{2}$$

For the rule  $x_2 = y_2$  and  $x_1 < y_1$  we have  $y_1$  possible combos, this can be shown by the closed form

$$\sum_{i=1}^{y_1} i = y_1$$

Combining these closed forms we get the closed form

$$R(y) = \frac{y_2(y_2 - 1)}{2} + y_1$$